

The key idea of the paper is to propose a cellular architecture that reduces the barrier to entry for new mobile operators by enabling users to consume access on-demand from any available operator. The authors do so by moving support for mobility and user management out of the network and into the end hosts. The motivation for the paper is to foster competition in the cellular market, as building a cellular network in the current landscape is capital intensive.

The authors propose CellBricks to tackle the problem, which deviates from the current cellular design in two major ways. It enables consumers to connect to and pay for service on demand from any infrastructure operator and it provides support for mobility. The paper argues that current cellular architecture is unsuitable for smaller scale providers because of two main reasons. Firstly, scaling coverage requires pre-established agreements and secondly, seamless mobility requires coordination between cell towers. Switching to a different provider disrupts TCP connections and degrades user experience.

A high level overview of operations of CellBricks involves on demand authentication and authorization, tamper proof and verifiable usage reporting by user equipment (UE) and telco (T1) to a broker telco B, and seamless switching for UE from T1 to T2. Such a design offers many benefits like removing scale bias, removing the need of pre-established contracts, simplifying cellular architecture, improving infrastructure efficiency and seamless integration of private networks with public networks. Cellbricks requires no change to Radio Access Network (RAN) and bTelcos can use any band available to them.

The authors discuss the design concerns of CellBricks and address the issues like secure attachments, seamless mobility and verifiable billing and QoS. CellBricks utilizes the existing radio layer connectivity, for authentication, CellBricks requires authentication between U and B, authentication between T and B and obtaining irrefutable proof that B authorized it to service U. For this purpose, SAP protocol is invoked. For seamless connectivity, CellBricks proposes a host driven approach which eliminates the concept of handovers. To handle IP changes, protocols like MCTCP and QUIC can be utilized, which allow the use of multiple underlying IP addresses for the same connection.

For QoS, CellBricks relies on independent traffic reports from both T and U, ensuring tamper proofing. Large or continuous discrepancies are considered as anomalies, based on which a reputation system is established for future cooperation. The experimental results show that the handover latency with CellBricks is comparable to the handover latency inherently present in the mobile networks.

While the work is interesting, and I am in favor of universal connectivity, competition and innovation, I do have some reservations on the work. Firstly, competition is necessary for consumer benefit but telcos benefit from monopoly, so they are less likely to be onboard with the solution. Secondly, tools exist to tamper secure partitions like IMEI and baseband on various devices (almost all Exynos and Snapdragon devices, which prevail the smartphone market). So tamper resistance cannot be fully realized. The paper also reminds me VolkFi, a smartphone that aimed to provide peer to peer connectivity. It was a similar concept to the one proposed in the paper, but it never got released and now it only exists on the wayback machine.